Literature map for mining the base operator L_λ and planning next steps

A self-contained recommendation memo

Context

This memo is written for the program centered on a symmetric base operator L_λ on a bounded logarithmic interval $I_\lambda = [-L, L]$, obtained from an energy decomposition of the restricted Weil quadratic form, and the perturbed operator

$$A_\lambda = L_\lambda + C_\lambda I + K_\lambda,$$

where $C_\lambda \in \mathbb{R}$ is a scalar shift and K_λ is finite-rank or otherwise very low complexity. The immediate questions are:

1. how to simplify and strengthen the current paper on L_λ;
2. how to organize it so that Dedekind zeta, Rankin-Selberg, and related L-functions can be treated with minimal rewiring;
3. how to derive new structural properties of L_λ;
4. how to obtain realistic routes toward the "simple-even" statement for A_λ;
5. and what other applications might reasonably follow from the same operator package.

The main recommendation of this memo is that the literature worth mining falls into two spines:

- a *functional-analytic spine*, which explains what the jump-form structure of L_λ already buys, how to package it cleanly, and which perturbative tools survive when one passes from L_λ to A_λ;
- an *explicit-formula / L-function spine*, which explains which features of the number-theoretic input are stable under passage from ζ to Dedekind zeta, automorphic, and Rankin-Selberg settings.

The two spines should not be merged too early. The best architecture is:

$$\text{abstract operator theorem} + \text{local verification of hypotheses from explicit formula data.}$$

That architecture is what makes the current paper both shorter and more extensible.

1. Executive triage: which literatures are actually worth mining?

Literature family	Priority	What it is best for	Why it matters here
Dirichlet forms and symmetric Markov processes	Very high	Markov property, irreducibility, invariance, positivity preservation, Beurling-Deny type packaging	This is the natural home of the jump-kernel rewrite of L_λ. It provides the cleanest language for the form domain, positivity, and the semigroup generated by the Friedrichs extension.
Positive / positivity-improving semigroups	Very high	Simplicity of the ground state, strict positivity, large-time behavior, semigroup criteria for uniqueness	If one can show positivity improvement or an equivalent irreducibility statement for the semigroup of L_λ, the simplicity of the bottom eigenvalue becomes conceptually straightforward.
Compact resolvent / compact embedding for form domains	Very high	Discreteness of spectrum, eigenbasis, quantitative eigenvalue growth inputs	Your compact-resolvent step is not merely technical; it is what turns the form into a fully spectral object. Mining this literature helps shorten the proof and makes extensions to weighted or less regular settings more robust.
Analytic perturbation theory and finite-rank perturbations	Very high	Stability of isolated eigenvalues, branch tracking, persistence of simplicity under perturbation	This is the correct language for moving from L_λ to A_λ once the base operator has been packaged correctly.
Nonlocal Schr"odinger / Feynman-Kac / intrinsic ultracontractivity	High	Ground-state shape, positivity, decay, regularity, domination, confining-potential analogies	Even though L_λ is not literally a standard fractional Schr"odinger operator, the analogy is close enough that this literature is useful for ground-state heuristics and for identifying transferable semigroup arguments.
Weil explicit formula for number fields and general L-functions	Very high	Abstracting what part of the proof is arithmetic and what part is operator theory	This is the key to making the paper transportable to Dedekind zeta and beyond.
Automorphic / Rankin-Selberg low-zero and Li-coefficient literature	High	General completed L-functions, gamma factors, self-duality issues, positivity surrogates	This literature clarifies which parts of the Connes-style framework should survive for more general L-functions, and which parts should be replaced by weaker symmetries than evenness.

Literature family	Priority	What it is best for	Why it matters here
Trace-formula literature tied to zero sums	Medium-high	Alternate positivity mechanisms and family-level interpretations	This is less immediately useful for the base paper, but especially relevant if the goal shifts from a single operator to families, averages, or generalized Connes-type approximations.

The least useful adjacent literatures are broad PDE regularity papers with no semigroup or form content, and generic spectral-geometry papers that do not touch positivity, compactness, or perturbation. For this project, elegant but non-operator-theoretic treatments of explicit formulas are usually less valuable than papers that package the explicit formula as a quadratic form or as a positivity statement.

2. The functional-analytic spine

2.1. Tier 1 references

The most immediately useful references are:

F1. Fukushima–Oshima–Takeda, *Dirichlet Forms and Symmetric Markov Processes.*

F2. Ouhabaz, *Analysis of Heat Equations on Domains.*

F3. Keller–Lenz–Vogt–Wojciechowski, *Note on basic features of large time behaviour of heat kernels.*

F4. Ben Amor, *Sobolev-Orlicz inequalities, ultracontractivity and spectra of time changed Dirichlet forms.*

F5. Kato, *Perturbation Theory for Linear Operators.*

These five already cover almost everything needed for the current base-operator paper: closed forms, Friedrichs extension, positivity preserving semigroups, irreducibility / positivity improvement, compact embedding, discrete spectrum, and perturbative continuation of isolated eigenvalues.

2.2. What each family buys

Dirichlet-form literature. This is the correct framework whenever the base form can be written as

$$\mathcal{E}_\lambda(f,f) = \iint_{I_\lambda \times I_\lambda} (f(u)-f(v))^2 J_\lambda(u,v)\,du\,dv + \text{lower-order terms},$$

with $J_\lambda \geq 0$ symmetric. In this language, the valuable outputs are:

- a canonical selfadjoint operator associated with the closed form;

- the Markov property and positivity preservation of the semigroup;
- irreducibility criteria expressed in terms of invariant sets or zero-energy decomposition;
- a standard route to the Friedrichs extension that is cleaner than ad hoc operator definitions.

For your paper, the main payoff is not merely conceptual elegance. It separates the proof into:

1. identify the principal jump form and verify the Dirichlet-form axioms;
2. treat scalar and finite-rank terms afterward.

That decomposition makes the extension to other L-functions feasible.

Positivity-improving semigroup literature. The Keller–Lenz–Vogt–Wojciechowski paper is especially useful because it distills what one really needs from positivity improvement: large-time convergence to ground states and a general mechanism turning positivity plus irreducibility into information about the bottom of the spectrum. In your setting, this literature matters for one very specific reason: if the semigroup generated by L_λ is positivity improving, then the lowest eigenvalue is automatically simple, and the associated eigenfunction can be chosen strictly positive. The later "even" conclusion can then be extracted from symmetry.

In other words, for the base operator the most valuable implication chain is

Dirichlet form $\Longrightarrow$ positive semigroup $\Longrightarrow$ irreducible or positivity improving $\Longrightarrow$ simple ground state.

This is more robust than proving simplicity by hand from Euler-Lagrange equations.

Compact embedding and compact resolvent literature. Ben Amor is valuable not because your current situation already requires an Orlicz-space detour, but because it gives a flexible template for proving discreteness of spectrum via compact embedding of the energy space into L^2. This becomes especially useful if later generalizations introduce weights, trace terms, nonstandard local parts, or weaker regularity at the boundary. For the present paper, the conceptual lesson is simple: once the form norm controls enough regularity or tightness on the bounded log interval, compact resolvent should be packaged as an embedding statement, not as a custom spectral argument.

Analytic perturbation and finite-rank perturbation theory. Kato is the indispensable reference when passing from L_λ to

$$A_\lambda = L_\lambda + C_\lambda I + K_\lambda.$$

The scalar shift is harmless. The real issue is the finite-rank term. The Kato viewpoint says: once the base eigenvalue is isolated, any perturbative argument should be phrased in terms of spectral projections, analytic branches, and Riesz integrals. This is the right language for proving persistence of simplicity along any parameterized deformation and for tracking how parity behaves when the perturbation commutes with reflection.

For your current goals, this literature should be mined in two modes:

- *soft mode*: preserve simplicity and evenness for small or continuous deformations;
- *hard mode*: derive a Birman-Schwinger or determinant criterion reducing changes in the bottom eigenvalue to a finite-dimensional matrix problem.

The second mode is especially attractive because K_λ is low-rank.

2.3. Secondary but useful references

A few additional references are worth mining selectively:

F6. Seifert–Wingert on perturbation of positive semigroups.

F7. Keller–Lenz–Wojciechowski on symmetries, reducing subspaces, and commuting operators.

F8. Arendt–Gl"uck on positive irreducible semigroups.

F9. Arendt–ter Elst–Gl"uck on strict positivity of principal eigenfunctions.

F10. Gesztesy and collaborators on Birman-Schwinger and finite-rank perturbative spectral analysis.

These are not replacements for the Tier 1 references. They are the papers to mine when you want sharper packaging or a cleaner argument for one specific step.

2.4. What to extract immediately for the current paper

The immediate extractions should be:

1. recast the main quadratic form as "principal Dirichlet-form part + bounded or finite-rank remainder";
2. state a short abstract theorem: *if a symmetric form on $L^2(I)$ is closed, Markovian, irreducible, and compactly embedded, then the associated operator has compact resolvent and simple ground state*;
3. verify those hypotheses once for the base operator;
4. move all number-theoretic work into the verification of the hypotheses.

This is the single most effective simplification strategy available.

3. The nonlocal Schrödinger / Feynman-Kac spine

3.1. Why this literature is relevant

At first glance, nonlocal Schrödinger papers are not about your operator. But they are often about exactly the features you care about: positivity of the semigroup, uniqueness and regularity of the ground state, intrinsic ultracontractivity, and how lower-order terms deform the spectral bottom. In particular, once L_λ is seen as a pure-jump symmetric operator on a bounded interval, the analogy with nonlocal Schrödinger operators with confining geometry becomes mathematically useful.

3.2. Best references to mine

The most useful papers in this lane are:

N1. Chen–Kim–Wang, *Intrinsic Ultracontractivity and Ground State Estimates of Non-local Dirichlet forms on Unbounded Open Sets.*

N2. Chen–Wang, *Intrinsic contractivity properties of Feynman-Kac semigroups for symmetric jump processes with infinite range jumps.*

N3. Baraniewicz, *Ground state decay for Schrödinger operators with confining potentials.*

3.3. How to use them

Their value is mostly strategic rather than literal.

- They supply semigroup criteria that often replace direct eigenfunction analysis.
- They suggest what hypotheses are natural if one wants sharper pointwise control on the ground state.
- They provide a menu of arguments for treating lower-order perturbations and for comparing operators via domination.

For your base paper, I would not import heavy pointwise ground-state technology unless you specifically want quantitative inequalities involving the ground state. But for future work, this literature is the best place to look if you want:

1. quantitative lower and upper bounds on the principal eigenfunction;
2. improved stability estimates under perturbation;
3. heat-kernel or semigroup inequalities leading to derived estimates for L_λ.

4. The explicit-formula / L-function spine

4.1. Tier 1 references

For portability to other L-functions, the most valuable references are:

A1. Weil on the explicit formula.

A2. Poitou for the number-field explicit formula viewpoint.

A3. Iwaniec–Kowalski, *Analytic Number Theory.*

A4. Lagarias, *Li coefficients for automorphic L-functions.*

A5. Büthe, *A method for proving the completeness of a list of zeros of certain L-functions.*

A6. de la Bretèche–Kuperberg, for a clean modern statement of Weil's explicit formula for Dedekind zeta and Artin L-functions.

A7. Battistoni–Molteni notes on explicit formulae for L-functions and class-group generators.

4.2. Why these are the right sources

Iwaniec–Kowalski. This is the basic infrastructure reference when you need completed L-functions, archimedean gamma factors, conductor normalizations, and standard automorphic analytic properties. It should be the default background citation in the generalized paper, especially for Rankin-Selberg and automorphic notation.

Lagarias. This paper is much more directly relevant to your program than its title suggests. It packages Weil positivity and Li coefficients for automorphic L-functions in a way that sits very close to Connes' perspective. For a generalized sequel paper, this is one of the best references for showing that your operator-theoretic constructions are not artifacts of the Riemann zeta function alone, but part of a larger automorphic "Weil form" ecosystem.

Büthe. This is extremely valuable because it formulates a rather general Weil-Barner explicit formula for a broad class of L-functions with explicit gamma data. For extension work, it provides a practical template for the abstract hypotheses one wants on the completed L-function before beginning any operator construction.

de la Bretèche–Kuperberg and Battistoni–Molteni. These are useful because they write the number-field explicit formula in a modern, computation-aware, and often cleaner form than older sources. For Dedekind zeta functions, this is where one can most quickly identify the exact archimedean kernels and finite-place sums that must be mirrored in the operator decomposition.

4.3. What to extract for generalization

For generalization purposes, the operator paper should be reorganized around the following abstract input data:

1. the completed L-function with its gamma factors;
2. the explicit formula written as zero sum = arithmetic sum + archimedean contribution + polar terms;
3. a test-function class stable under Mellin convolution and truncation to $[\lambda^{-1}, \lambda]$;
4. an archimedean kernel decomposition with positive or at least controllable principal part;
5. a clear identification of which residual terms are scalar, bounded, compact, or finite-rank.

That list should become a front-end "black box" theorem in any generalized sequel.

5. Dedekind zeta, automorphic, and Rankin-Selberg settings

5.1. Dedekind zeta functions

Dedekind zeta is the most natural next target because the passage from ζ to ζ_K preserves the basic Weil-explicit-formula geometry while changing the archimedean part in a controlled and intelligible way through the signature (r_1, r_2). This means:

- the operator-theoretic strategy should survive with only moderate changes;
- the main new work is in handling multiple archimedean contributions and the precise normalization of the scalar term;
- the bounded-log-interval strategy still makes sense;
- finite-place terms remain accessible through the same Mellin-convolution formalism.

In practical terms, Dedekind zeta is the correct proving ground for a generalized version of the current paper.

5.2. Automorphic L-functions

For self-dual automorphic L-functions, the program still looks plausible, but one should expect a shift in emphasis:

- positivity of the principal jump kernel may survive only after a more delicate decomposition;
- the natural symmetry may be self-duality rather than literal evenness on the line;
- the number of gamma factors grows, so the archimedean operator may become a sum of several positive kernels plus a more complicated lower-order correction.

This suggests that the generalized paper should not hard-code "evenness" too early. Instead it should formulate a broader conclusion:

> uniqueness of the ground state in the symmetry class singled out by the functional equation.

Only in the Riemann zeta case does this collapse to an even ground state in the literal reflection sense.

5.3. Rankin-Selberg

Rankin-Selberg is more difficult, but conceptually richer. The main reasons are:

1. the gamma factors are more numerous and more asymmetric;

2. the arithmetic side may naturally encode convolution data that do not obviously produce a scalar-valued positive kernel in the same way as the zeta case;
3. family-level positivity and trace-formula positivity may become more relevant than single-operator positivity.

This does *not* make Rankin-Selberg a bad target. It only means that two different styles of arguments may be needed:

- an operator-theoretic argument for the base selfadjoint piece;
- a trace-formula or finite-dimensional perturbative argument for the residual terms.

The recent work of Wong is worth mining precisely because it relates zero sums of Rankin-Selberg L-functions to trace-formula expressions. That is the right literature if the single-operator framework begins to creak and one needs a family-level or spectral-side positivity input.

6. Goal-by-goal recommendations

6.1. Goal 1: improve the current paper for simplicity, elegance, robustness, and readability

Primary recommendation

Refactor the paper so that the main theorem is deduced from one abstract operator proposition and one explicit-formula proposition.

How to do it

1. Replace long direct calculations by a front-loaded theorem of the form:

   > If the principal energy is a closed irreducible Dirichlet form on $L^2(I_\lambda)$ with compact embedding of the form domain, then the associated operator has compact resolvent and simple ground state.

2. Prove a separate explicit-formula theorem showing that the archimedean term admits the required positive jump-kernel decomposition and that all residual terms are scalar, bounded, or finite-rank.
3. Relegate notation-heavy manipulations and alternate derivations to appendices.

Literature to mine first

Fukushima–Oshima–Takeda, Ouhabaz, Ben Amor, Kato.

Expected payoff

This should shorten the logical spine of the paper substantially and make external verification much easier. It also makes the proof visibly modular.

6.2. Goal 2: structure the paper so that Dedekind zeta, Rankin-Selberg, and others become direct extensions

Primary recommendation

Introduce an abstract "completed L-function package" section early in the paper, even if the present paper only instantiates it for ζ.

What this package should contain

1. completed L-function and gamma factors;
2. functional equation and symmetry class;
3. explicit formula on a Mellin-convolution test space;
4. decomposition of the archimedean contribution into positive kernel part plus lower-order remainder;
5. classification of the remainder terms by operator type.

Why multiple proof styles may be needed

For Dedekind zeta functions, the same style of argument should plausibly work after local modifications. For general automorphic or Rankin-Selberg L-functions, the archimedean decomposition may no longer yield one single transparent positive kernel. In that case, the proof may require a hybrid approach:

- abstract operator theory for the principal positive part;
- a symmetry-class argument for the functional-equation involution;
- perturbation theory or finite-dimensional reduction for the non-positive remainder.

Literature to mine first

Iwaniec–Kowalski, Lagarias, Büthe, de la Bretèche–Kuperberg, Battistoni–Molteni.

6.3. Goal 3: strengthen the results within the paper and derived inequalities

Most realistic strengthening targets

The low-hanging fruit is not a dramatic theorem jump, but a package of sharper structural consequences:

1. monotonicity of the lowest eigenvalue under natural positive deformations;
2. quantitative spectral-gap lower bounds whenever an irreducibility estimate can be upgraded;
3. stability bounds for the lowest eigenpair under finite-rank perturbation;
4. improved inequalities for $\mu_0(\lambda)$ by variational trial functions built from the jump-kernel structure;
5. semigroup-derived inequalities, such as positivity-preserving comparisons or domination bounds.

Best literature for this goal

Kato for perturbative differentiability and eigenvalue branches; positive-semigroup papers for spectral gap and simplicity consequences; nonlocal Schrödinger literature for ground-state comparison heuristics.

What is probably not low-hanging fruit

Trying to extract very sharp pointwise asymptotics of the ground state from the current structure is probably premature. So is trying to derive fine Weyl-law asymptotics unless the compact embedding argument is already fully optimized.

6.4. Goal 4: derive new properties of L_λ that may be useful in any context

Best targets

The most reusable new properties are the ones that are insensitive to the arithmetic origin of the operator:

1. positivity preservation and, if possible, positivity improvement of the semigroup;
2. compact resolvent and stability of the form domain under symmetry;
3. continuity or analyticity of the lowest eigenpair in auxiliary deformation parameters;
4. Birman-Schwinger reduction for the finite-rank perturbation;
5. operator monotonicity for natural families of truncated kernels.

Why these matter

These statements can be reused whether the eventual target is Connes Step 1, a general automorphic variant, a de Bruijn–Newman style deformation, or a family-averaged application.

6.5. Goal 5: outline a concrete plan toward proving simple-even for A_λ

The recommended plan

The cleanest route is a three-layer plan.

Layer 1: solve the base problem abstractly. Show that L_λ has compact resolvent and simple ground state using Dirichlet-form irreducibility plus compact embedding.

Layer 2: isolate symmetry. Show that the reflection operator $Jf(u) = f(-u)$ commutes with L_λ and with the perturbation K_λ, hence with A_λ. Once the ground state is unique, it must lie in a one-dimensional J-invariant eigenspace, hence be even or odd; positivity excludes oddness, so it is even.

Layer 3: pass to the perturbed operator. Use either:

(a) a direct positivity-improving argument for the perturbed semigroup if available; or

(b) a finite-rank perturbation argument showing that the lowest isolated simple eigenvalue of the base operator persists and remains in the even sector.

Why this is preferable to attacking simple-even for A_λ all at once

Because it isolates the two genuinely different mechanisms:

- simplicity comes from positivity and irreducibility;
- evenness comes from symmetry plus uniqueness.

When those are separated, the proof becomes more transportable.

Where multiple proof techniques may be needed

If the perturbation K_λ does not preserve positivity in a way that yields a positivity-improving semigroup directly, one may have to combine:

1. an unperturbed positivity proof;
2. a Kato continuation argument in a coupling parameter t for $L_\lambda + tK_\lambda$;

3. a finite-dimensional determinant or Birman-Schwinger analysis to prevent eigenvalue crossing at the bottom.

This is likely the correct hard-analysis route to Connes Step 1.

6.6. Goal 6: identify other applications beyond the Alain Connes program

Most plausible applications

The most plausible applications are those in which one wants a selfadjoint nonlocal operator extracted from an explicit formula, rather than the full RH program itself:

1. Li-criterion and Weil-positivity reinterpretations for broader classes of L-functions;
2. deformations of the explicit-formula operator in the style of de Bruijn–Newman questions;
3. family-level or averaged operators related to low-lying zeros;
4. explicit inequalities for conductors, first zeros, or residues where the operator-theoretic viewpoint supplies monotonicity or comparison principles;
5. numerical and computer-assisted spectral experiments on generalized Weil operators.

Why this matters strategically

These applications provide mathematically meaningful outputs even if the full Connes approximation hypothesis remains out of reach. They also make the theory more publishable as a standalone operator-theoretic framework.

7. Where one argument will probably not suffice

A recurring theme is that different L-functions force different proof architectures. The following dichotomy is important.

Case A: one positive kernel dominates

This is the zeta and probably Dedekind-zeta regime. Here one expects a single principal Dirichlet form plus bounded or finite-rank corrections. The natural proof style is:

$$\text{Dirichlet form} \to \text{compact resolvent} \to \text{simple ground state} \to \text{symmetry}.$$

Case B: several local pieces compete or positivity is only partial

This is the more likely automorphic / Rankin-Selberg regime. Here one should expect to need:

- a decomposition into several operator pieces;
- sector or symmetry decomposition before proving uniqueness;
- perturbative or finite-rank arguments rather than a direct positivity-improving semigroup argument;
- possibly family-level positivity coming from the trace formula instead of single-operator positivity.

Therefore the generalized paper should be designed from the outset to allow hybrid proofs.

8. A prioritized reading and extraction plan

Phase I: mine immediately for the current paper

1. Fukushima–Oshima–Takeda: extract the minimum closed-form / irreducibility / invariance package you need.
2. Ouhabaz: extract the semigroup language and any irreducibility criterion you want to state cleanly.
3. Keller–Lenz–Vogt–Wojciechowski: extract the positivity-improving-to-ground-state mechanism.
4. Ben Amor: extract the compact-embedding-to-discrete-spectrum packaging.
5. Kato: extract the perturbative continuation statements for isolated simple eigenvalues.

Phase II: prepare the extension architecture

1. Büthe: abstract hypothesis list for admissible completed L-functions.
2. de la Bretèche–Kuperberg and Battistoni–Molteni: concrete Dedekind-zeta formulas.
3. Iwaniec–Kowalski and Lagarias: automorphic normalization and Weil positivity framework.

Phase III: attack A_λ

1. formulate $A_\lambda(t) = L_\lambda + C_\lambda I + tK_\lambda$;
2. prove that the lowest eigenvalue branch stays isolated on the relevant range of t;
3. prove symmetry preservation and exclude odd ground states;
4. if needed, reduce the no-crossing problem to a finite-dimensional matrix criterion using the rank of K_λ.

9. Concrete recommendations, distilled

If one wants a compact answer to "what kinds of papers are actually worth mining?", the answer is:

1. **Dirichlet-form papers and books:** these are the most valuable for the current base-operator proof itself.
2. **Positivity-improving semigroup papers:** these are the most valuable for simplicity of the ground state.
3. **Compact-resolvent / compact-embedding papers:** these are the most valuable for making the spectral packaging short and robust.
4. **Kato and finite-rank perturbation papers:** these are the most valuable for moving from L_λ to A_λ.
5. **Explicit-formula papers written in modern abstract language:** these are the most valuable for extension to Dedekind zeta and automorphic settings.
6. **Nonlocal Schrödinger and Feynman-Kac papers:** these are the most valuable for sharpening ground-state and semigroup statements, but they are secondary to the first four categories.

If time is limited, mine the literature in exactly that order.

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